

# KodetoCareer — Implementation Plan

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Version 1.0 | 10-Week MVP Roadmap

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## 1. Team Structure

### Recommended Team for MVP

Role | Count | Responsibilities

Backend Developer (Node.js) | 1–2 | API development, database, integrations

Frontend Developer (React.js) | 1 | Admin panel and college portal

Flutter Developer | 1 | Mobile app (iOS + Android)

UI/UX Designer | 1 | Figma designs, prototypes, handoff

DevOps / Infrastructure | 0.5 (part-time or backend dev doubles) | AWS setup, CI/CD, deployment

Product Manager / Founder | 1 | Requirement clarification, QA sign-off

QA Tester | 1 (from Week 6) | Test execution, bug reporting

**Total: 5–7 people**

If budget is constrained:

- A senior full-stack developer can handle both Node.js backend and React frontend
- Flutter developer is a separate skill set — do not compromise here
- UI/UX can be done by founder using the detailed brief and Figma templates initially

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## 2. Pre-Sprint Preparation (Week 0 — Before Development Starts)

### Infrastructure & Tooling Setup (DevOps + Backend):

- [ ] AWS account setup with IAM roles and billing alerts
- [ ] Create: S3 buckets, RDS PostgreSQL instance (t3.micro for dev), ElastiCache Redis
- [ ] Configure CloudFront distribution
- [ ] Set up GitHub repository with branch strategy (main, develop, feature/\*)
- [ ] Configure GitHub Actions CI pipeline (lint + test on every PR)

- [ ] Set up staging and production environments
- [ ] Domain registration and SSL certificates (AWS Certificate Manager)
- [ ] Configure SES for transactional email, MSG91 for SMS
- [ ] Set up Sentry for error tracking (free tier)

**Design Preparation:**

- [ ] Finalize brand guidelines (logo, colours, fonts) — use this brief
- [ ] Set up Figma workspace with design system components
- [ ] Design Priority 1 screens (see UI/UX Brief Section 12)
- [ ] Create Figma prototype for student onboarding flow for early user testing

**Planning:**

- [ ] Break each epic into Jira/Linear tickets with acceptance criteria
- [ ] Create API contract (OpenAPI YAML) before any coding starts
- [ ] Set up shared Postman collection for API testing
- [ ] Define test data: 2 colleges, 5 batches, 50 students for seed data

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## 3. Sprint Plan (10 Weeks × 1-Week Sprints)

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### SPRINT 1 — Foundation & Auth (Week 1)

**Backend:**

- [ ] Set up Express.js project structure with TypeScript
- [ ] Configure Prisma with PostgreSQL, run initial migration
- [ ] Implement users table, all auth endpoints:
  - POST /auth/register
  - POST /auth/verify-email
  - POST /auth/login
  - POST /auth/forgot-password
  - POST /auth/reset-password
  - POST /auth/refresh
  - POST /auth/logout
- [ ] JWT middleware (access token validation)
- [ ] Role-based access control middleware
- [ ] Rate limiting setup

- [ ] Basic error handling middleware

**Flutter:**

- [ ] Project setup (folder structure, dependencies in pubspec.yaml)
- [ ] Set up Riverpod state management
- [ ] Set up Dio HTTP client with interceptors
- [ ] Implement: Splash, Onboarding, Login, Register, Email OTP, Profile Setup screens
- [ ] Secure token storage (flutter\_secure\_storage)
- [ ] Navigate to Dashboard on successful login

**React (Admin):**

- [ ] Project setup with Vite, Tailwind, shadcn/ui
- [ ] Set up React Router with protected routes
- [ ] Admin login page
- [ ] Super Admin dashboard shell (with sidebar navigation)
- [ ] Axios instance with auth interceptors

**Sprint 1 Deliverable:** A user can register, verify email, complete profile, and log in to the app. Admin can log in to the web panel.

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## SPRINT 2 — College & User Management (Week 2)

**Backend:**

- [ ] College CRUD endpoints
- [ ] Trainer CRUD endpoints
- [ ] College Admin creation (with auto-invite email)
- [ ] Student bulk import (CSV parsing, validation, response)
- [ ] Student profile endpoints (GET, PUT)
- [ ] Audit log middleware (auto-log all write operations)

**React (Admin):**

- [ ] College Management page (list, create, edit, deactivate)
- [ ] College detail page with tabs (Overview, Students, Batches, Reports)
- [ ] Trainer management page
- [ ] Student management page with search, filter

**Flutter:**

- [ ] My Profile screen (view and edit)
- [ ] Profile photo upload

- [ ] Settings screen (change password, notifications)

**Sprint 2 Deliverable:** Admin can create colleges, trainers, and import students. Students can view and edit their profiles.

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## SPRINT 3 — Course & Content Structure (Week 3)

### Backend:

- [ ] Course CRUD with publishing workflow
- [ ] Module CRUD (with sort order)
- [ ] Lesson CRUD (with all lesson types)
- [ ] Lesson Notes (PDF) upload to S3
- [ ] S3 presigned URL generation for secure content delivery
- [ ] Student lesson progress tracking endpoints

### React (Admin):

- [ ] Course Management page (create, edit, publish)
- [ ] Course builder UI (module + lesson drag-and-drop order)
- [ ] Lesson editor (tabbed: video source, PDF upload, text content)
- [ ] File upload with progress indicator

### Flutter:

- [ ] Course list (Learn tab)
- [ ] Course detail (modules accordion)
- [ ] Lesson detail (tabs: Video, Notes, My Notes)
- [ ] PDF viewer (flutter\_pdfview)
- [ ] Personal notes editor

**Sprint 3 Deliverable:** Admin can create complete courses with content. Students can browse course content and view PDFs.

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## SPRINT 4 — Video Delivery (Week 4)

### Backend:

- [ ] Large video upload via S3 multipart (presigned parts)
- [ ] AWS MediaConvert job creation Lambda (or triggered from Node.js)
- [ ] MediaConvert completion webhook → update lesson video status

- [ ] Adaptive video URL serving (HLS via CloudFront)
- [ ] Video progress tracking (resume position)

**Flutter:**

- [ ] Video player integration (video\_player + chewie)
- [ ] Custom controls (speed, quality, fullscreen, PiP)
- [ ] Video progress persistence (last position saved)
- [ ] "Mark complete" button logic (appears at 80%)
- [ ] Module/lesson completion celebration animation (Lottie)

**React (Admin):**

- [ ] Video upload progress UI with status tracking
- [ ] Video processing status indicator on lesson card

**Sprint 4 Deliverable:** Full video lesson delivery working end-to-end on mobile.

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## SPRINT 5 — Batch & Attendance (Week 5)

**Backend:**

- [ ] Batch CRUD endpoints
- [ ] Batch student enrollment (individual + CSV bulk)
- [ ] Class session CRUD
- [ ] Attendance marking endpoints (with 24h override rule)
- [ ] Attendance summary calculation
- [ ] Attendance export (Excel/PDF generation with ExcelJS + PDFKit)

**React (Admin):**

- [ ] Batch Management page (create wizard, list, detail)
- [ ] Batch student list (enroll, remove, status)
- [ ] Trainer: Attendance marking screen
- [ ] Attendance history and edit UI

**Flutter:**

- [ ] Student: View attendance per batch (percentage, session-by-session)
- [ ] Attendance history screen
- [ ] Push notification receipt for attendance marked

**Sprint 5 Deliverable:** Trainers can create sessions and mark attendance. Students see their attendance. Admin sees per-batch reports.

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## SPRINT 6 — Assessments (Week 6)

### Backend:

- [ ] Quiz CRUD (with question bank)
- [ ] Quiz attempt start/auto-save/submit
- [ ] Quiz scoring engine (MCQ\_SINGLE, MCQ\_MULTIPLE, TRUE\_FALSE)
- [ ] Quiz leaderboard computation
- [ ] Assignment CRUD
- [ ] Assignment submission + grading endpoints
- [ ] Notification triggers for quiz results and graded assignments

### React (Admin):

- [ ] Quiz builder (question types, options, correct answer, marks)
- [ ] Assignment creation and management
- [ ] Grade assignments (view submission, enter marks, feedback)
- [ ] Quiz analytics (per-question correct % breakdown)

### Flutter:

- [ ] Quiz list (Tests tab)
- [ ] Quiz detail and start flow
- [ ] Quiz screen (single question, timer, auto-save)
- [ ] Quiz result screen (score, answer review)
- [ ] Assignment list, submission flow
- [ ] Quiz leaderboard screen

**Sprint 6 Deliverable:** Full assessment lifecycle working — creation, attempt, scoring, and results.

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## SPRINT 7 — Certificates & Documents (Week 7)

### Backend:

- [ ] Certificate eligibility check endpoint
- [ ] Certificate PDF generation (ReportLab/PDFKit with template)
- [ ] Certificate upload to S3 and delivery URL
- [ ] Public verification endpoint (no auth)
- [ ] Bulk certificate generation (Bulk queue)
- [ ] Student documents upload (offer letters, internship letters)

- [ ] Placement record CRUD

**React (Admin):**

- [ ] Certificate management page (eligibility table, override, generate)
- [ ] Bulk generation progress UI (socket-powered live updates)
- [ ] Upload offer/internship letter to student profile
- [ ] Placement record management

**Flutter:**

- [ ] My Certificates screen (list, view, share)
- [ ] Certificate PDF viewer
- [ ] Share sheet integration (LinkedIn, WhatsApp, copy link)
- [ ] My Documents screen (offer letters, download)

**Sprint 7 Deliverable:** Certificate generation pipeline complete. Students can view and share certificates.

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## SPRINT 8 — Placement Hub & Job Board (Week 8)

**Backend:**

- [ ] Placement analytics endpoints
- [ ] Job opportunity CRUD
- [ ] Job interest recording
- [ ] Placement readiness score calculation (composite algorithm)
- [ ] Placement status update

**React (Admin):**

- [ ] Placement analytics dashboard (charts: placed by month, by company, avg package)
- [ ] Post job opportunity form
- [ ] Job management (edit, deactivate)
- [ ] Student placement status management

**Flutter:**

- [ ] Placement Hub screen (status, score, quick actions)
- [ ] Job board (list, filter by type)
- [ ] Job detail + "I'm Interested" flow
- [ ] Placement status display

**Sprint 8 Deliverable:** Placement tracking and job board live. Admin can see placement analytics.

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## SPRINT 9 — Communication, College Portal & Analytics (Week 9)

### Backend:

- [ ] Announcement send endpoint (scoped to all/college/batch/individual)
- [ ] Notification recipient creation (fan-out)
- [ ] FCM push notification integration
- [ ] Email notification system (all templates from TRD Section 10)
- [ ] SMS integration (MSG91 for OTP and alerts)
- [ ] College admin scoped endpoints
- [ ] Analytics aggregation endpoints
- [ ] Weekly report auto-generation (Bull cron job)

### React (Admin):

- [ ] Communication center (compose announcement, scope selector, send)
- [ ] College admin portal (dedicated routes, college-scoped views)
- [ ] Analytics centre (platform-wide charts)
- [ ] Student at-risk report

### Flutter:

- [ ] Notification centre (in-app notification list, mark read)
- [ ] Push notification tap handling + deep link navigation

**Sprint 9 Deliverable:** Communication system live. College admins can log in and view their dashboards.

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## SPRINT 10 — Polish, Testing & Launch Prep (Week 10)

### Quality Assurance (all team):

- [ ] End-to-end test all user flows (student, trainer, admin, college admin)
- [ ] Load testing (simulate 500 concurrent users on API)
- [ ] Mobile testing on real devices (minimum: Android mid-range, Android budget, iPhone 12)
- [ ] API security audit (OWASP top 10 check)
- [ ] Data validation testing (CSV imports with malformed data)

### Performance:

- [ ] Audit and fix slow API endpoints (anything > 500ms under load)
- [ ] Implement Redis caching for high-traffic read endpoints
- [ ] Image and asset compression audit
- [ ] Flutter app size optimisation (tree shaking, deferred libraries)



**Polish:**

- [ ] All empty states designed and implemented
- [ ] All error messages are human-readable (no raw API error codes exposed)
- [ ] Skeleton loaders on all content screens
- [ ] Offline handling on Flutter (graceful error when no network)
- [ ] Review all notification copy

**Deployment:**

- [ ] Production infrastructure finalised
- [ ] Environment variables secured in AWS Secrets Manager
- [ ] Database seeded with pilot data (3 colleges, real course content)
- [ ] App Store and Play Store submission (Flutter)
- [ ] DNS configuration, SSL verification
- [ ] Monitoring dashboards live (CloudWatch, Sentry)
- [ ] Runbook written for common operational tasks

**Pilot Onboarding:**

- [ ] 3 pilot college agreements signed
- [ ] College admins trained (1-hour walkthrough per college)
- [ ] First student batch imported and notified
- [ ] Trainer walkthrough for attendance and quiz creation

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## 4. Risk Register

Risk | Likelihood | Impact | Mitigation

Video processing pipeline delayed | Medium | High | Build with YouTube embed first, own upload in Sprint 4

Flutter developer bottleneck | Medium | High | Prepare detailed Flutter widget specs before Sprint 1

College onboarding slower than expected | High | Medium | Don't wait for all 3 — go live with 1 college for alpha

MediaConvert costs exceed budget | Low | Medium | Cap video uploads at 720p initially, monitor spend

Push notifications not delivered on budget Android | Medium | Medium | Test FCM early (Sprint 1), have SMS fallback

AWS RDS pricing | Low | Low | Start with t3.micro, upgrade on need

College admin resistance to adoption | Medium | High | Design simplest possible college portal for MVP

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5. Post-MVP Roadmap (Version 2 — Months 4–7)

| Feature                                             | Priority | Estimated Effort |
|-----------------------------------------------------|----------|------------------|
| Resume builder                                      | P1       | 3 weeks          |
| AI mock interview (Claude API)                      | P1       | 4 weeks          |
| Coding assessment platform (Monaco Editor + Judge0) | P1       | 5 weeks          |
| WhatsApp automation (via Interakt)                  | P2       | 2 weeks          |
| Recruiter portal (separate login)                   | P2       | 4 weeks          |
| College analytics advanced charts                   | P2       | 2 weeks          |
| Gamification (points, badges, leaderboard)          | P3       | 3 weeks          |
| Multi-language support (Hindi)                      | P3       | 3 weeks          |
| ---                                                 |          |                  |

6. Budget Estimate (MVP)

Development Cost (10 weeks)

| Role                                | Rate          | Duration   | Cost      |
|-------------------------------------|---------------|------------|-----------|
| Backend Dev (Node.js)               | ■60,000/month | 2.5 months | ■1,50,000 |
| Flutter Dev                         | ■70,000/month | 2.5 months | ■1,75,000 |
| React Dev                           | ■55,000/month | 2.5 months | ■1,37,500 |
| UI/UX Designer                      | ■50,000/month | 1.5 months | ■75,000   |
| QA Tester                           | ■35,000/month | 1 month    | ■35,000   |
| **Total Dev Cost**   **~■5,72,500** |               |            |           |

Infrastructure Cost (Monthly, Post-Launch)

| Service                             | Estimated Cost |
|-------------------------------------|----------------|
| AWS EC2 (2× t3.medium)              | ■12,000        |
| AWS RDS PostgreSQL (db.t3.small)    | ■6,000         |
| AWS S3 + CloudFront (1TB/month)     | ■4,000         |
| AWS MediaConvert (100 videos/month) | ■3,000         |

AWS SES (10,000 emails) | ■500

Redis (ElastiCache t3.micro) | ■3,000

Domain + SSL | ■2,000

Firebase (FCM push) | Free

MSG91 SMS (1000 OTPs) | ■800

Sentry (error tracking) | Free

**\*\*Total Monthly\*\* | \*\*~■31,300/month\*\***

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*End of Implementation Plan*